

Skills Summary

Multiplying Fractions and Mixed Numbers

Use this reference while completing your worksheet or when studying.

Skill 1 — Fraction times a whole number

Rule: multiply the whole number by the numerator and keep the denominator: the whole number counts how many copies of the fraction you take. If the answer is bigger than 1, rewrite it as a mixed number.

Example: Multiply $\frac{3}{4} \times 8$.

Step 1. Take 8 copies of three fourths — multiply 8 by the numerator and keep fourths.

Step 2. $\frac{3 \times 8}{4} = \frac{24}{4} = 6$

Try it: Multiply $\frac{2}{3} \times 12$.

check: 8

Skill 2 — Fraction times a fraction

Rule: multiply the numerators together and the denominators together, then simplify:
 $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$.

Example: Multiply $\frac{2}{3} \times \frac{3}{5}$.

Step 1. Tops together and bottoms together, then look for a common factor to simplify.

Step 2. $\frac{2 \times 3}{3 \times 5} = \frac{6}{15}$, and dividing top and bottom by 3 gives $\frac{2}{5}$

Try it: Multiply $\frac{3}{4} \times \frac{2}{5}$.

check: $\frac{6}{10}$

Skill 3 — Multiplying mixed numbers

Plan: rewrite every mixed number as an improper fraction *first*, multiply as in Skill 2, then change the answer back to a mixed number.

Example: Multiply $1\frac{1}{2} \times 2\frac{1}{3}$.

Step 1. Rewrite both as improper fractions before multiplying anything.

Step 2. $1\frac{1}{2} = \frac{3}{2}$ and $2\frac{1}{3} = \frac{7}{3}$, so $\frac{3}{2} \times \frac{7}{3} = \frac{21}{6} = \frac{7}{2} = 3\frac{1}{2}$

Try it: Multiply $2\frac{1}{2} \times 1\frac{1}{5}$.

check: 3

(!) Watch out: never multiply the whole parts and fraction parts separately. $1\frac{1}{2} \times 2\frac{1}{3}$ is *not* $2\frac{1}{6}$ — rewrite as improper fractions first.